

Body Cosmos VR

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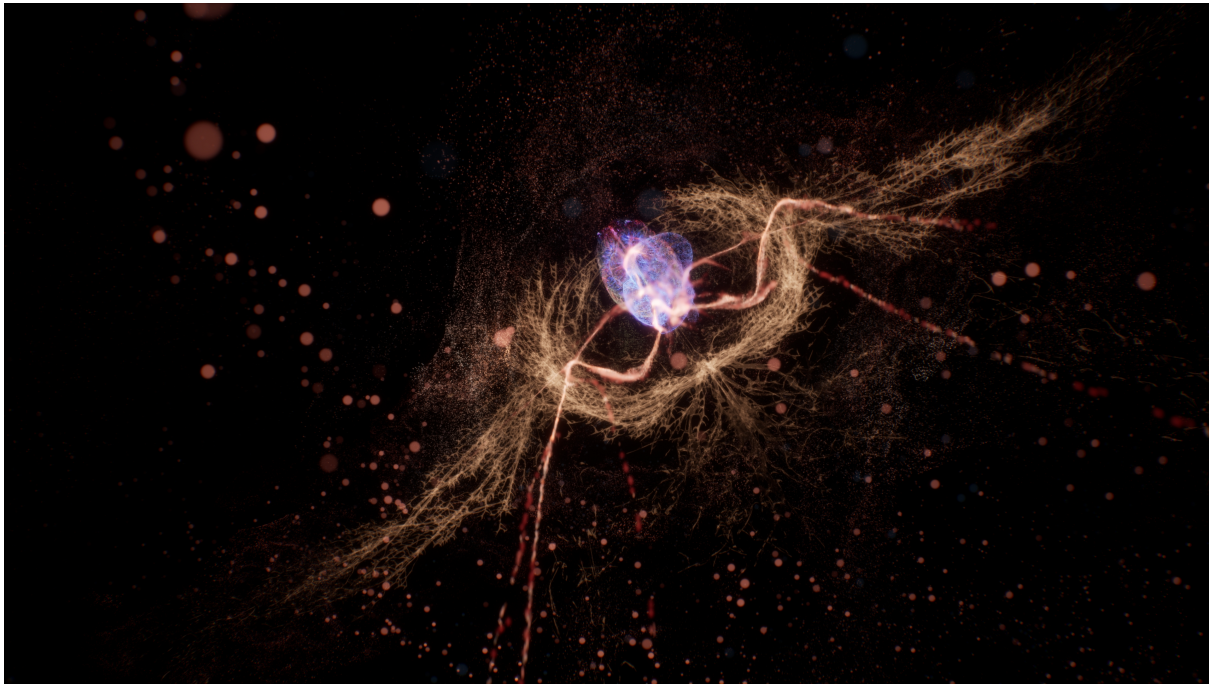


Figure 1: Screenshot of Unreal Engine’s real-time demo (© Rungu Lin 2023).

ABSTRACT

Body Cosmos is an immersive and interactive artwork that connects the human body with the cosmic environment through real-time bio-data. Using volumetric rendering techniques and particle system, we create a surreal virtual reality that reflects the intricate structures of human anatomy and celestial nebulae. We integrate a heart rate sensor and an EEG headband to capture and process bio-data into emotion indicators, which influence the visualization of particles in the artwork. This creates an intimate, personal connection with the cosmos, transcending immediate presence and nurturing a perpetual presence within the cosmic expanse. We also provide two engagement modes: Roller Coaster Mode and Free Explore Mode, allowing different levels of exploration. Body Cosmos is more than a visual spectacle; it sparks curiosity, expands the

imagination, and enhances awareness of our embodied and cosmic identity.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Visualization*; *Interaction design*; • **Applied computing** → *Arts and humanities*.

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“Body Cosmos” is an artwork that explores the interconnectedness of the human body in a digital cosmic environment (Figure 2). We create a surreal virtual reality that reflects the intricate structures of human anatomy and celestial nebulae, learning from volumetric rendering techniques. This immersive experience connects the macrocosm of the universe with the microcosm of human life, revealing the enduring resonance of our biological rhythms in the cosmic dance (Figure 1). “Body Cosmos” integrates real-time bio-data gathered through heart rate monitors and EEG devices. This life data influences the visualization, creating an intimate, personal

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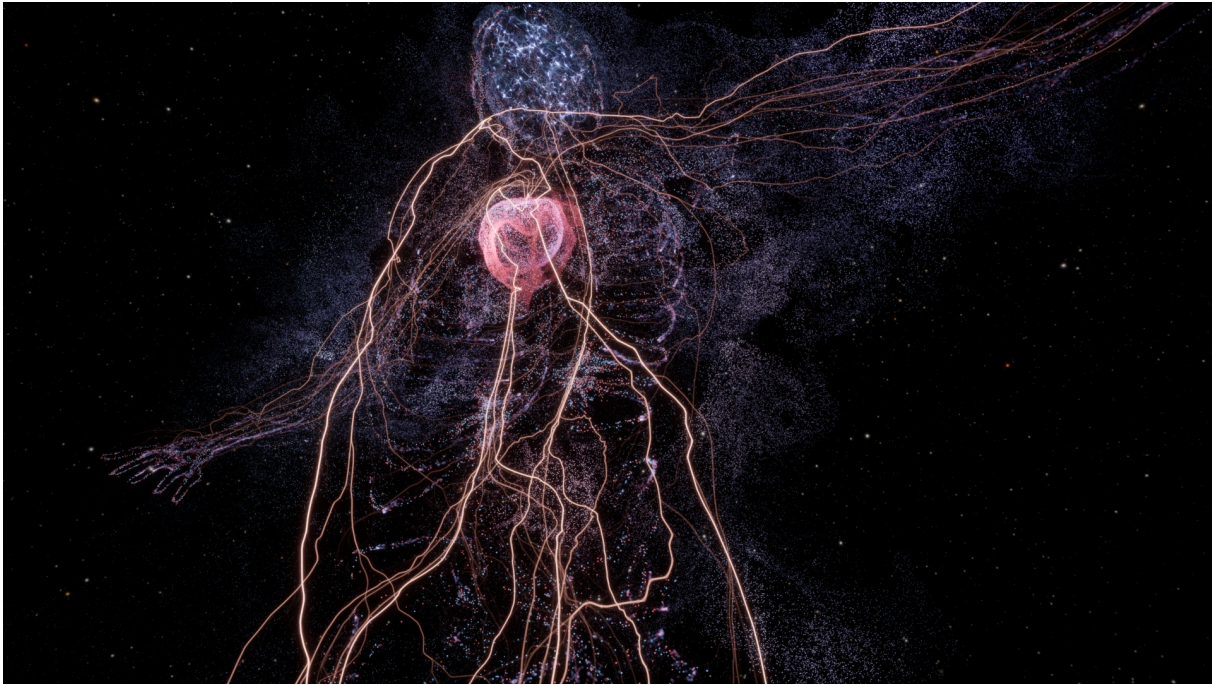


Figure 2: Screenshot of Unreal Engine's real-time demo (© Rungu Lin 2023).(© Rungu Lin 2023).

connection with the cosmos. The artwork transcends immediate presence, reflecting on the intertwined relationships that humans can form with digital data, thus nurturing a perpetual presence within the cosmic expanse.

Based on DICOM data from NSD dataset[Allen et al. 2022], we create an immersive and interactive experience driven by users' real-time bio-data. Particles represent the interconnection between individuals and the universe. The conversion of DICOM data into image sequences and further manipulation in Houdini and Unreal Engine 5 allows the depiction of particles driven by custom shaders responsive to real-time bio-data. In collaboration with Flowtime[Inc. 2023], we employ heart rate sensors and EEG devices to capture and process bio-data into emotion indicators, thereby adjusting various particle parameters in the visualization. To ensure an immersive experience, we provide two engagement modes: Roller Coaster Mode, a guided tour through the Body Cosmos, and Free Explore Mode, allowing independent exploration.

"Body Cosmos" is more than a visual spectacle. It is grounded on historical perspectives[Allers 1944][Lester 2012][Sagan 1980] and sparks curiosity, expands the imagination, and enhances awareness of our perpetual presence in the universe. This project explores three manifestations of the human body in relation to technology: the cyberbody[Haraway 2000], virtual body[Hayles 2000], and bio-body. Each of these representations expands our understanding of embodiment[Hallis 1999] and identity[Lasko-Harvill 1992], presenting a new perspective on the human body's relationship with the cosmos.

As we look ahead, we intend to extend Body Cosmos to include a wider range of bio-data sensors and explore possible psychological and physiological benefits that such immersive bio-interactive

experiences might offer. Furthermore, we aspire to promote social interactivity by developing multi-user capabilities, transforming this into a shared cosmic journey. In essence, Body Cosmos is an exploration of self within the vast cosmic tapestry, highlighting our perpetual presence in the universe we inhabit.

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